Job Loss in the Presence of Informality: Evidence from Brazil

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November 6, 2026

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Abstract

I study how recovery after job displacement varies with local informality in Brazil. Combining administrative employment records for 2000–2017 with regional informality measures from the 2000 Census, I estimate event studies of formal employment, earnings, hours, and hourly wages. Displaced workers in the event-study sample return to formal employment within five years. Preliminary estimates show persistent earnings and hours losses, with larger initial losses in formal earnings and employment in higher-informality regions. Among workers with observed positive formal earnings and hours, higher informality is associated with smaller hourly-wage penalties. The persistence of regional differences varies across outcomes and specifications. A two-sector search model illustrates how more frequent informal offers can increase selectivity over formal jobs and raise accepted wages, while leaving the effect on time to formal return ambiguous. Worker composition, pre-displacement trends, and selection into formal return limit a causal interpretation of the regional comparisons. The findings distinguish changes in formal employment and hours from wage changes among workers observed in formal jobs; they do not measure losses in total labor income.

1 Introduction

After losing a formal job, a worker's prospects depend on the opportunities available in the surrounding labor market. Where informal employment is widespread, work outside the formal sector may provide a source of income during adjustment. Those opportunities may also change which formal jobs workers are willing to accept and how quickly they return. Understanding recovery after job loss therefore requires examining both workers' attachment to formal employment and the wages they receive when re-employed.

How do the formal employment and earnings consequences of job displacement vary across local labor markets with different levels of informality? I study this question in Brazil, combining matched employer–employee administrative records from RAIS for 2000–2017 with regional informality measures from the 2000 Census. The administrative records allow me to follow workers' formal employment, earnings, hours, and hourly wages before and after displacement. The Census measures the prevalence of informal employment in each microregion, providing a way to compare recovery across local labor markets within the same national institutional setting.

I estimate event studies around employer-initiated separations between 2006 and 2011, comparing displaced workers with workers who are never displaced. The analysis focuses on workers with established employment relationships, restricting displaced workers to those who return to formal employment within five years. I first estimate average post-displacement paths, then compare regions in the highest and lowest quartiles of informality and relate separately estimated regional effects to the regional informality measure. I also examine non-return in the broader cohort of eligible displaced workers. These exercises concern formal-sector outcomes: an absence from RAIS does not reveal whether a worker is informally employed, unemployed, or outside the labor force, and the data do not measure earnings from informal work.

The estimates show large declines in formal earnings and hours following displacement, with shortfalls persisting five years after the event. Initial losses are larger in regions with higher informality. In the year after displacement, the estimated formal-employment shortfall is 22.7 percentage points in the highest-informality quartile and 15.5 percentage points in the lowest, relative to their respective comparison-group trajectories. The earnings losses in these two groups subsequently converge, while the continuous regional analysis finds a persistent association between higher informality and larger losses in earnings and hours relative to workers' pre-displacement benchmarks. The persistence of the regional differences thus depends on the outcome and specification.

Hourly wages show a different pattern. Among workers with observed positive formal earnings and hours, wage penalties are smaller in higher-informality regions. In the year after displacement, the estimated log hourly-wage effect is −0.013 in the highest-informality quartile and −0.094 in the lowest. The continuous regional estimates also associate higher informality with smaller wage penalties after the displacement year. In the broader displaced cohort, higher informality is associated with a larger share of workers who have no subsequent formal-employment record through the end of observation. Taken together, these patterns highlight the importance of distinguishing formal employment and hours from wages among workers observed in formal jobs.

The estimates are preliminary. Displaced and comparison workers differ substantially in sector composition, with public administration particularly prevalent among the comparison workers, and some outcomes exhibit differences in pre-displacement trends. Regional informality also covaries with other local labor-market characteristics. These features limit a causal interpretation of the estimates, and the regional comparisons do not isolate the effect of an exogenous change in informality. Selection into formal return and into positive earnings in a given year also matters for interpreting the wage results.

To organize possible mechanisms, I develop a two-sector job-search model in which unemployed workers receive formal and informal offers. Holding other primitives fixed, more frequent informal offers raise the reservation wage, weakly lower formal entry from unemployment, and weakly raise the mean accepted formal wage. Informal employment interrupts formal search in the model. The effect on time to formal return remains ambiguous: greater selectivity reduces formal acceptance but also avoids some informal spells. The model provides a qualitative account of how informal opportunities can affect job acceptance. It does not identify the mechanism behind the regional patterns or establish that higher accepted wages imply smaller displacement-related wage losses.

Related Literature

The paper connects research on job displacement with research on informal labor markets. Studies of job displacement document persistent earnings losses and examine why their magnitude and sources differ across workers and labor markets (Jacobson et al., 1993; Davis and von Wachter, 2011). Stevens (1997) highlights the contribution of repeated job losses to this persistence. Lachowska et al. (2020) distinguish earnings, hours, and hourly wage losses, finding an important role for persistent wage-rate reductions. Schmieder et al. (2023) show that moves to lower-paying

employers account for much of the wage losses following displacement in Germany and help explain their variation over the business cycle. Using a harmonized design across European countries, Bertheau et al. (2023) show that differences in subsequent employment are central to cross-country differences in displacement losses. These findings motivate examining several dimensions of recovery and the local opportunities available to displaced workers.

Search models explain why finding another job need not restore a worker's previous earnings. Krolikowski (2017), Jung and Kuhn (2019), and Jarosch (2023) emphasize job ladders and differences in employment stability as sources of persistent displacement losses. This perspective motivates considering how the opportunities available during search affect both the wages and continuity of subsequent employment.

A related literature studies search and adjustment in economies with informal employment. Meghir et al. (2015) develop an equilibrium search model with formal and informal firms, estimated using Brazilian data, to study wages, sector allocation, and labor-market policy. Informal employment also provides a margin of adjustment to adverse labor-demand shocks. In Brazil, Dix-Carneiro and Kovak (2019) document the expansion of informal employment during the longer-run adjustment to trade liberalization. Ponczek and Ulyssea (2022) find larger overall employment losses following liberalization in regions with stricter enforcement of labor regulations. In an estimated equilibrium model, Dix-Carneiro et al. (2026) show how informal employment can cushion the unemployment response to adverse shocks even as aggregate real income falls.

This paper examines a related question at the level of individual job displacement. Its contribution is to document how subsequent formal employment, hours, and wages vary with pre-existing regional informality, and to provide a search-based interpretation of their joint patterns. The empirical analysis measures adjustment within formal employment; it does not assess the overall income protection provided by informal work. [1]

Section 2 describes the data and empirical strategy. Section 3 presents the results and their limitations. Section 4 develops the search framework and its implications, and Section 5 concludes.

[1] Rogerson and Schindler (2002) study the welfare costs of displacement when workers cannot fully insure against earnings risk, highlighting the role of insurance and self-insurance in translating earnings losses into welfare costs.

2 Data and Empirical Strategy

The main data source is RAIS (*Relação Anual de Informações Sociais*), a set of matched employer–employee administrative records maintained by Brazil's Ministry of Labor. I use records for 2000–2017 to follow workers' formal employment histories. The data provide information on workers' age, educational attainment, gender, and earnings, as well as contracted hours, employment dates, average monthly remuneration, and reasons for separation. The records also identify each establishment and report its size and industry. Employers are required to report employment information annually and face penalties for noncompliance. The main limitation of RAIS is its coverage of formal employment: when a worker is absent from the records, I cannot distinguish unemployment from informal employment or nonparticipation in the labor force.

The second data source is the 2000 Brazilian Demographic Census. It complements RAIS with information on employment inside and outside the formal sector, allowing me to characterize each microregion's labor market. Although the Census does not allow me to follow workers over time, it distinguishes formal employment, informal employment, unemployment, and nonparticipation. I classify employees without a signed work card as informally employed (Dix-Carneiro and Kovak, 2019). I also classify self-employed workers who do not contribute to the public social security system as informally employed (Engbom et al., 2022). Workers do not face penalties for reporting informal employment, which supports the use of Census responses to measure regional informality.

2.1 Panel Construction and Sample Selection

I remove observations with missing identifiers and exclude any worker whose records contain a contract with zero average wages, rather than dropping only the affected contract. I restrict the sample to workers aged 25–55 and exclude workers who move across microregions from the baseline sample. These migrants account for a small share of workers.

I construct an annual panel from the employment contract records. Formal earnings and hours worked are set to zero when an individual has no observed formal employment during an entire year. I classify a worker as formally employed in a given year if their annual formal earnings are strictly positive. Job tenure is defined as the number of months of continuous employment with the same employer.

I aggregate earnings and hours across employment contracts to construct annual outcomes. For

each worker-year, I construct annual earnings by summing monthly remuneration multiplied by the duration of each employment contract within that year. I construct annual contracted hours analogously, converting weekly contracted hours to monthly hours using four weeks per month. I divide both annual totals by twelve to obtain monthly averages over the full calendar year. These averages therefore incorporate periods without formal employment within the year. I identify displacement as employer-initiated termination, with or without just cause (RAIS separation codes 10 and 11). A binary indicator records whether a worker experiences such a termination in a given year.

The event studies focus on displacements between 2006 and 2011, inclusive. To enter the treatment group, workers must have at least one displacement in this window at which their tenure is at least 36 months. This requirement follows the focus on workers with established employment relationships in Davis and von Wachter (2011). I restrict the event-study sample to workers who return to formal employment within five years of displacement. A related restriction is used by Albrecht (2022), who requires workers to find a full-time job within three years. I analyze workers who return after a longer absence or never return to formal employment separately. The control group comprises workers who are never displaced and who also satisfy a minimum tenure requirement of 36 months.

Following Stevens (1997), I use each treated worker's first displacement within the study window as the reference event. Subsequent displacements are measured as an additional outcome. I follow workers for at least five years before and five years after the reference displacement. For the baseline sample, I randomly draw 10% of eligible treated and control workers. After panel preparation, the baseline sample contains 187,540 treated workers and 291,280 controls. For the regional estimates and the high/low informality comparisons, I use a sample stratified by region, containing 193,865 treated workers and 366,348 controls. The high/low analysis further restricts this sample to workers in the lowest and highest quartiles of regional informality: 137,599 treated workers and 190,596 controls in the lowest quartile, and 6,014 treated workers and 65,200 controls in the highest. These counts describe the prepared panels; the observations used in estimation can differ across outcomes.

Table 1 summarizes worker characteristics one year before the actual or assigned event. In the baseline sample, treated workers are slightly younger and have lower monthly earnings, lower hourly pay, and shorter recorded tenure than controls, despite working more contracted hours. Controls are also more likely to have completed tertiary education and are much more concentrated in public administration, defence, and social security: 50.3% of controls versus 3.1% of treated workers. Similar

Table 1: Worker characteristics before the event

	Baseline sample		Stratified sample	
	Treated	Controls	Treated	Controls
Workers	187,540	291,280	193,865	366,348
Age (years)	36.8	38.1	36.8	38.0
	(6.1)	(5.5)	(6.1)	(5.5)
Monthly formal earnings (2010 BRL)	1,243	2,030	1,233	1,750
	(1,534)	(2,496)	(1,515)	(2,316)
Monthly contracted hours	172.0	157.2	172.1	154.5
	(41.0)	(40.9)	(41.6)	(41.7)
Hourly pay (2010 BRL)	7.41	13.91	7.37	12.15
	(12.85)	(20.05)	(14.50)	(18.81)
Recorded tenure (months)	78.3	134.8	78.4	133.5
	(49.5)	(69.2)	(49.4)	(68.8)
Female (%)	38.6	49.0	38.6	53.4
Completed tertiary education (%)	9.9	25.2	9.8	23.2
Public administration sector (%)	3.1	50.3	3.4	58.1
Positive formal earnings (%)	98.6	99.0	98.6	98.9

Notes: Standard deviations appear below continuous means; percentages are indicated. Characteristics refer to one year before the actual or assigned event, except gender and education, which use each worker's modal category over the observed history. All workers have a reference-year row. Counts describe prepared panels; estimation samples can differ by outcome. Both samples use equal worker weights; stratified columns cover all quartiles. Earnings and hours are annual totals divided by twelve, including zeros. Hourly pay requires positive earnings and hours. Tenure is the maximum across reference-year contracts, not necessarily tenure in the displaced job. Hourly-pay and tenure means exclude missing observations (about 1.0–1.4%). Tertiary education includes completed undergraduate and postgraduate degrees. Public administration includes defence and social security (CNAE 95 division 75); percentages use all workers, including those without a recorded sector. Source: RAIS, 2000–2017.

differences appear in the stratified sample.

Differences in employment stability across sectors raise concerns about comparability. These estimates are preliminary; a planned revision will assess their sensitivity to excluding public administration, defence, and social security.

The regional analysis uses microregions, merging some units to maintain consistent boundaries over the study period. I measure regional informality as the number of informally employed workers, classified as described above, divided by the region's total labor force. I interpret this measure as an indicator of workers' exposure to informal employment opportunities relative to formal ones. There are 558 microregions with informality levels ranging from 21.5% to 88.9% and a median of 55.6%.

Figure 1 shows the geographic distribution of regional informality.

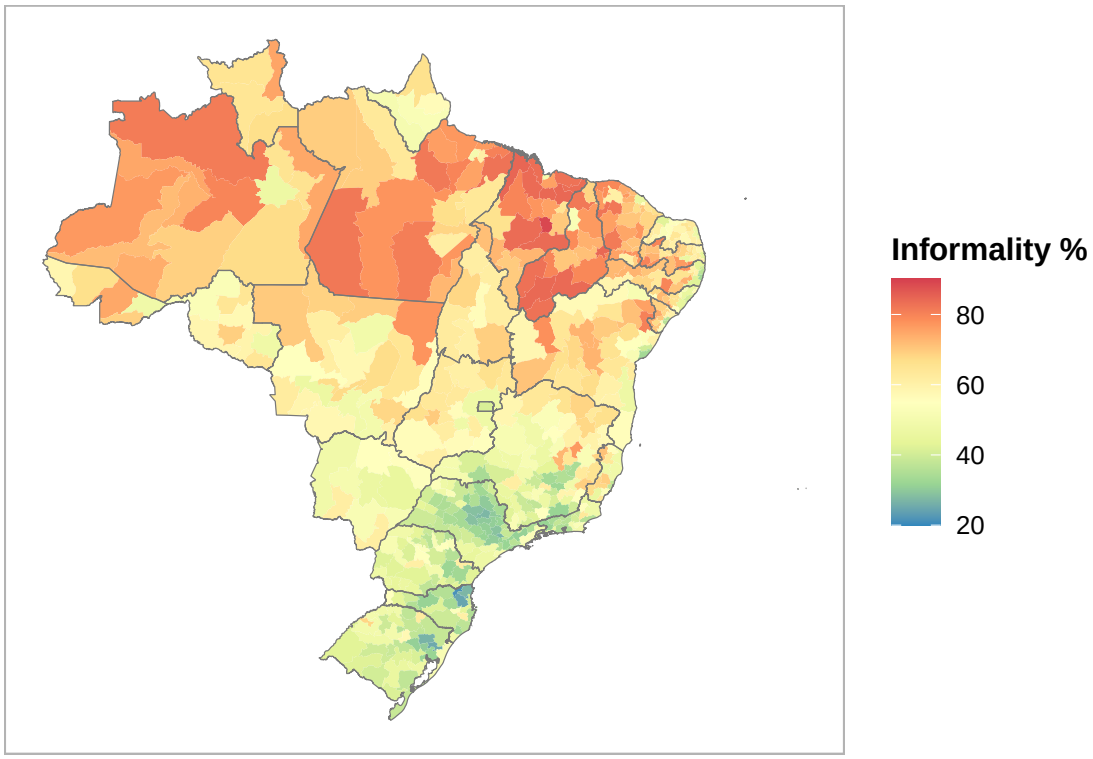


Figure 1: Regional informality in Brazil, 2000

Notes: Informality is the share of the labor force classified as informally employed in each microregion, expressed as a percentage. Calculations use Census sampling weights. Source: 2000 Brazilian Demographic Census.

2.2 Empirical Strategy

I estimate earnings losses using the following event-study specification:

$$y_{it} = \alpha_i + \gamma_t + X'_{it}\beta + \sum_{\substack{k=-6 \\ k\neq -1}}^{6} \delta_k D_{it}^k + e_{it}. \tag{1}$$

Following Lachowska et al. (2020), I include individual and calendar-year fixed effects, denoted here by α_i and γ_t, respectively, to account for persistent differences across workers and common aggregate shocks. In the baseline and high/low informality specifications, X_{it} includes age squared, gender and education indicators, and gender- and education-specific age slopes to allow for different life-cycle profiles. Average pre-displacement hours and earnings are each interacted with calendar-year indicators to allow for differential outcome trends.

I also include state (UF) fixed effects, reference-sector-by-year fixed effects, and establishment-size-category-by-year fixed effects. Reference sector is measured at the two-digit CNAE level and held fixed over time: for treated workers, it is the sector recorded for a displaced job in their first displacement year within the study window; for controls, it is the sector in their assigned reference year. Establishment size is measured annually.[2] The separate regressions for each microregion omit education-specific age slopes, reference-sector-by-year effects, and state fixed effects.

For $k \neq -6, 6$, the indicator D_{it}^k equals one if worker i was displaced in period $t-k$, and zero otherwise. The endpoint indicators pool observations outside the interior event window: D_{it}^6 equals one six or more years after displacement, while D_{it}^{-6} equals one six or more years before displacement. I omit the endpoint coefficients from the figures because these pooled periods have a different interpretation from the individual years in the interior of the window.

The coefficients of interest, δ_k for $k > 0$, trace the average effect of job loss on the outcome k years after displacement. A causal interpretation requires parallel trends between the treatment and control groups: in the absence of displacement, their outcomes would have followed parallel paths conditional on the included controls and fixed effects.

I next examine whether the initial impact of displacement and the subsequent recovery differ between regions with low and high informality. Let Z_i indicate that worker i works in a region with high informality. I estimate the following specification:

[2] Within each worker-year, I use the establishment-size category from the contract with the earliest end date, treating ongoing contracts as ending on December 31. The measure is carried forward in years without observed formal employment and can change when a worker moves to a different employer.

$$y_{it} = \alpha_i + \gamma_t + X'_{it}\beta + Z_i\zeta + \sum_{\substack{k=-6 \\ k\neq -1}}^{6} \delta_k D_{it}^k + \sum_{\substack{k=-6 \\ k\neq -1}}^{6} \eta_k D_{it}^k Z_i + e_{it} \tag{2}$$

For $k \geq 0$, the displacement effect relative to the year before displacement is δ_k in low-informality regions and $\delta_k + \eta_k$ in high-informality regions; η_k measures the high-minus-low difference at event time k.

3 Results

Job displacement is followed by a sharp decline in formal earnings and hours, with losses persisting five years after the event. Regional informality is associated with the size and persistence of these losses, but the relationship differs across outcomes. Workers in high-informality regions experience larger initial losses in earnings, hours, and formal employment. The continuous cross-region estimates also show a persistent negative association between informality and the displacement effects on earnings and hours measured relative to workers' pre-displacement benchmarks. Hourly pay follows a different pattern: among workers with observed formal earnings and hours, higher informality is associated with a smaller post-displacement wage penalty.

The distinction between formal earnings and total earnings is central to the interpretation. RAIS records formal employment, so an absence from the data may represent informal employment, unemployment, or withdrawal from the labor force. Moreover, the displaced workers in the event studies are restricted to those who return to formal employment within five years. The estimates therefore describe formal-sector outcomes for this selected group. I first present the average effects of displacement, then examine differences by regional informality, and finally relate these patterns to non-return and subsequent displacement.

3.1 Earnings, hours, and employment after displacement

Figure 2 reports the effects of displacement on formal earnings and hours, each divided by the worker's own pre-displacement average. The benchmarks are constructed from years at least two years before displacement, and the event-study coefficients are normalized to the year immediately preceding displacement. Years without formal employment enter these outcomes as zeros. The estimates consequently capture both changes in formal employment and changes in earnings and



Figure 2: Displacement effects on relative formal earnings and hours

Notes: Outcomes are divided by each worker's pre-displacement mean, computed using years at least two years before displacement. Years without formal employment enter as zeros. Displaced workers return to formal employment within five years; comparison workers are never displaced. Coefficients are relative to year −1. Shaded bands are pointwise 95% confidence intervals based on standard errors clustered by worker. The binned endpoints of the estimation window are omitted from the figure.

hours during years with formal work.

Relative earnings fall by 0.447 benchmark units in the displacement year and by 0.577 in the following year. Most of the subsequent recovery occurs by year two. Five years after displacement, the earnings shortfall remains 0.274 benchmark units. Hours follow a similar path: the estimated losses are 0.413 in the displacement year, 0.436 in year one, and 0.190 in year five. These are differences relative to the comparison group's trajectory, expressed in units of each worker's own benchmark. For example, the year-one earnings coefficient represents a shortfall of 57.7 percentage points of that benchmark; it is not a 57.7% decline relative to counterfactual earnings in that year.

The employment estimates help explain why the losses deepen in the year after displacement. Positive formal earnings at some point during the year define formal employment here; continuous employment throughout the year is not required. The estimated employment effect is positive in the displacement year (1.9 percentage points), when displaced workers have already held a formal job, but falls to −16.0 percentage points in year one. The shortfall narrows to 8.8 percentage points in year two and 4.0 percentage points in year five. Returning to formal work within five years thus does not imply uninterrupted formal employment over the subsequent period.

Losses also remain substantial when the sample in each year is restricted to workers with positive

formal earnings. Conditional relative earnings fall by 0.483 benchmark units in the displacement year and 0.416 in year one, remaining 0.223 below the comparison trajectory in year five. Conditional hours fall by 0.442 on impact and remain 0.153 below the comparison trajectory in year five. These estimates show that years entirely without formal employment do not account for all of the earnings and hours losses. They do not, however, isolate outcomes among continuously employed workers: a worker with positive earnings may still spend much of the year outside formal employment.

The log outcomes in Figure 3 provide another view of this adjustment, excluding zero observations. Log earnings decline by 0.588 log points in the displacement year and remain 0.279 log points below the comparison trajectory in year five. Log hours fall by 0.608 on impact and remain 0.209 below the comparison trajectory in year five. From year one onward, the hourly-wage penalty is much smaller and changes little, ranging from approximately 0.08 to 0.07 log points. The recovery in log earnings over this period is therefore accompanied primarily by a recovery in hours. Hours are constructed from contracted hours and job duration, so this margin combines the duration of formal employment with hours associated with the jobs held during the year.

Hourly wages rise in the displacement year before falling in year one. This short-lived increase warrants caution, particularly because the benchmark-relative wage measure is sensitive to unusually large ratios. I therefore place more weight on the sustained post-displacement pattern than on the year-zero wage coefficient. Throughout, the log and conditional estimates concern the workers observed with the relevant positive outcomes in each year, whose composition can change after displacement.

3.2 Differences between high- and low-informality regions

I next compare microregions in the top and bottom quartiles of regional informality, measured using the 2000 Census. Quartiles are defined across regions. Each path in Figure 4 is an estimated displacement effect relative to the comparison workers in that group; the difference between paths is a difference in estimated effects, rather than a difference in the levels of earnings or employment across regions.

The initial disruption is larger in high-informality regions. In year one, relative earnings are 0.709 benchmark units below the comparison trajectory in the highest-informality quartile, against 0.563 in the lowest quartile. The corresponding losses in hours are 0.534 and 0.424. The high-informality group subsequently recovers more of its initial earnings loss: by year five, the relative-

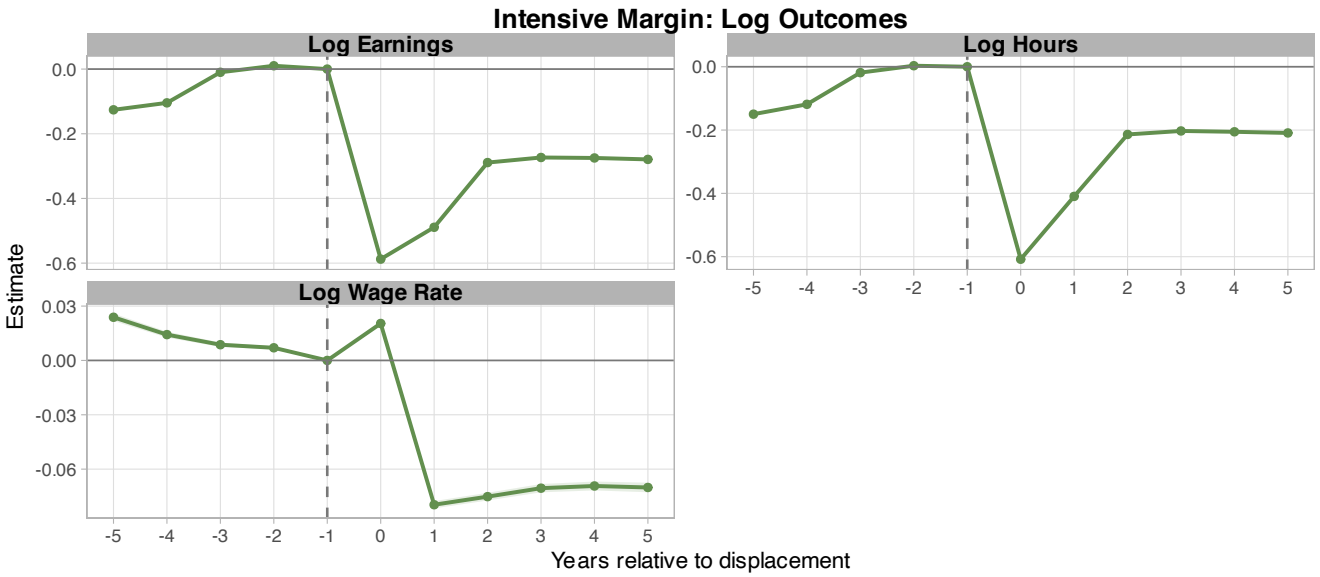


Figure 3: Displacement effects on log formal earnings, hours, and hourly wages

Notes: Log outcomes exclude observations with zero or otherwise invalid values for the relevant outcome. Hourly wages are formal earnings divided by hours. The sample restriction, normalization, and inference are as in Figure 2. Positive annual earnings do not imply employment throughout the year.

earnings coefficients are similar, at −0.263 and −0.271, respectively. The hours shortfall remains somewhat larger in the high-informality group (0.203 versus 0.182). Thus, the larger initial loss does not translate into a uniformly larger earnings loss at every later horizon.

The employment paths show a larger shortfall in high-informality regions as well. Formal employment falls by 22.7 percentage points in year one in the highest quartile, compared with 15.5 percentage points in the lowest. By year five, the respective shortfalls are 5.1 and 3.8 percentage points. Both groups recover substantially, but the high-informality group retains a larger employment loss.

Among workers with observed positive outcomes, the wage comparison goes in the opposite direction. In year one, the log hourly-wage effect is −0.013 in the high-informality group and −0.094 in the low-informality group; by year five, the effects are −0.013 and −0.080. Log earnings also show a smaller loss in the high-informality group at later horizons. These comparisons must be read alongside the differences in the pre-displacement paths, particularly for wages. The post-displacement gap cannot be cleanly attributed to displacement if the groups were already on different relative trajectories.

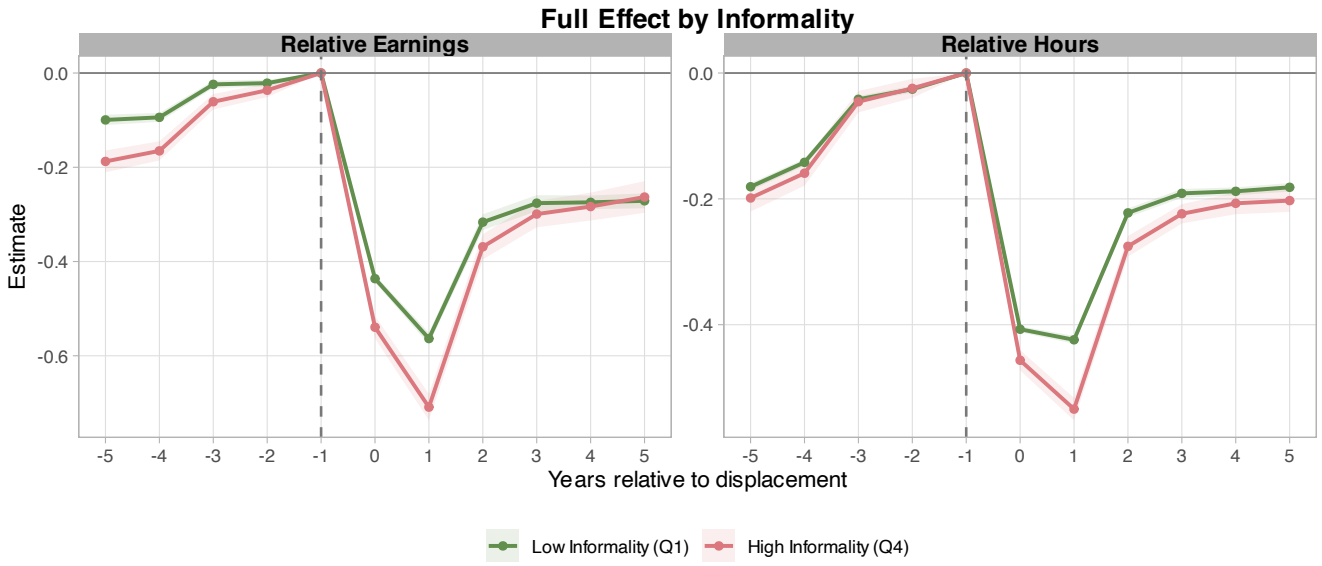


Figure 4: Relative earnings and hours by regional informality

Notes: High and low informality denote the top and bottom quartiles of the distribution across microregions. Each curve reports a displacement effect relative to never-displaced workers, normalized to year −1. High-group confidence intervals account for the covariance between the main event-time coefficients and their high-informality interactions. Bands are pointwise 95% confidence intervals based on worker-clustered standard errors.

3.3 The continuous regional informality gradient

The quartile comparison summarizes the extremes of the informality distribution. To use the variation across all regions, I estimate the event study separately within each microregion and then relate the estimated displacement effects to regional informality. For each outcome and event time k, the second step is

$$\hat{\delta}_{rk} = \alpha_k + \theta_k \, \text{Inf}_r + \gamma_k \, \text{Pre}_{rk} + u_{rk}. \tag{3}$$

Here Inf_r is the regional informality share and Pre_{rk} summarizes the slope of the region's pre-displacement event-study coefficients. The sample contains 527 regions with at least 100 eligible displaced workers. Regressions use inverse-variance weights, $1/\hat{\sigma}^2_{rk}$, and heteroskedasticity-robust standard errors. More precise regional estimates therefore contribute more to the fitted relationship. The weights vary across outcomes and horizons and do not give every region equal influence. [3]

A negative θ_k means that higher informality is associated with a more negative displacement effect. Because informality is measured as a share from zero to one, a 10 percentage point difference

[3]For post-displacement periods, the pretrend control uses the pre-displacement coefficients. For a pre-displacement period, that period's coefficient is excluded when constructing its control. This avoids using the same coefficient directly on both sides of the regression.

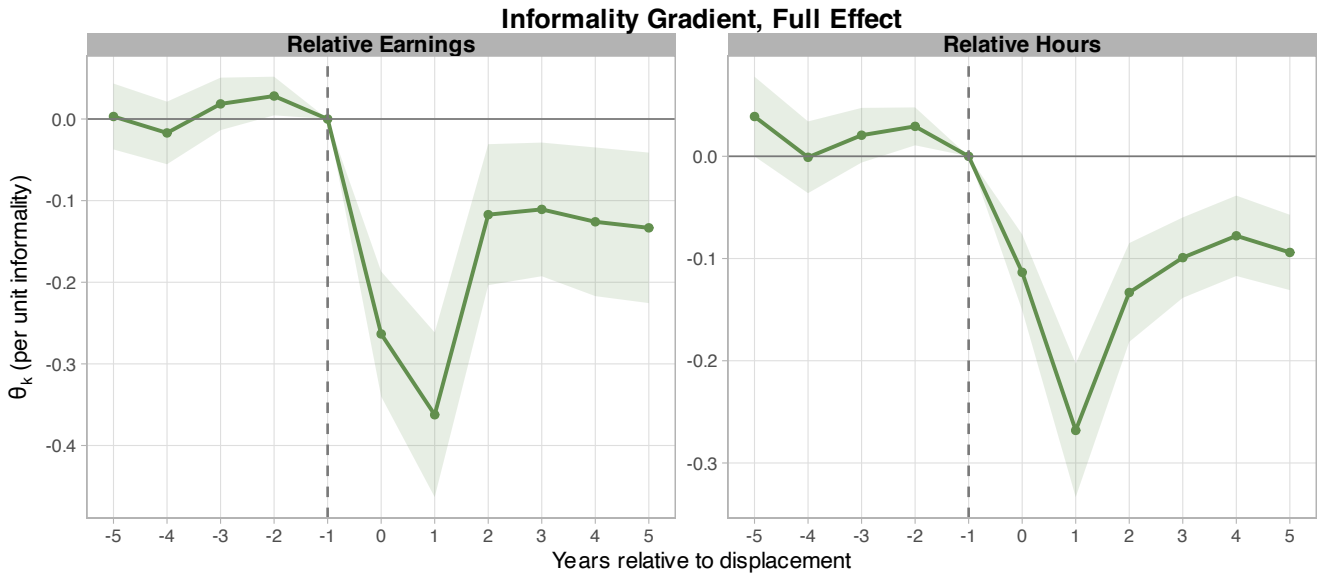


Figure 5: Regional informality gradients in relative earnings and hours

Notes: Curves report θ_k from equation (3), using inverse-variance weights and a regional pretrend control. Informality is measured from zero to one; multiply a coefficient by 0.1 for a 10 percentage point difference in informality. Shaded bands are pointwise 95% confidence intervals using heteroskedasticity-robust standard errors. The reference-period coefficient is normalized to zero.

in informality corresponds to a difference of $0.1\theta_k$ in the estimated effect. The specification conditions on the pretrend summary but does not adjust for regional education or labor-market size. Its coefficients describe regional associations rather than an independently identified causal effect of informality.

Figure 5 shows negative gradients for relative earnings and hours from the displacement year through year five. For earnings, θ_k is -0.263 on impact, -0.362 in year one, and -0.133 in year five. A 10 percentage point increase in informality is thus associated with an additional earnings shortfall of 3.6 percentage points of the pre-displacement benchmark in year one and 1.3 percentage points in year five. The corresponding hours gradients are -0.268 and -0.094, implying additional shortfalls of 2.7 and 0.9 percentage points of the hours benchmark. Both gradients are statistically different from zero at the 5% level throughout years zero to five.

These estimates support a persistent association between higher informality and larger formal earnings and hours losses. The agreement with the quartile comparison is strongest for the initial losses and for hours. For later earnings, the quartile paths converge whereas the continuous gradient remains negative. The comparisons use different specifications and give regions different influence: the two-step approach fits a continuous relationship to separately estimated regional effects, while the quartile comparison pools workers in the two extreme groups. They should therefore be reported

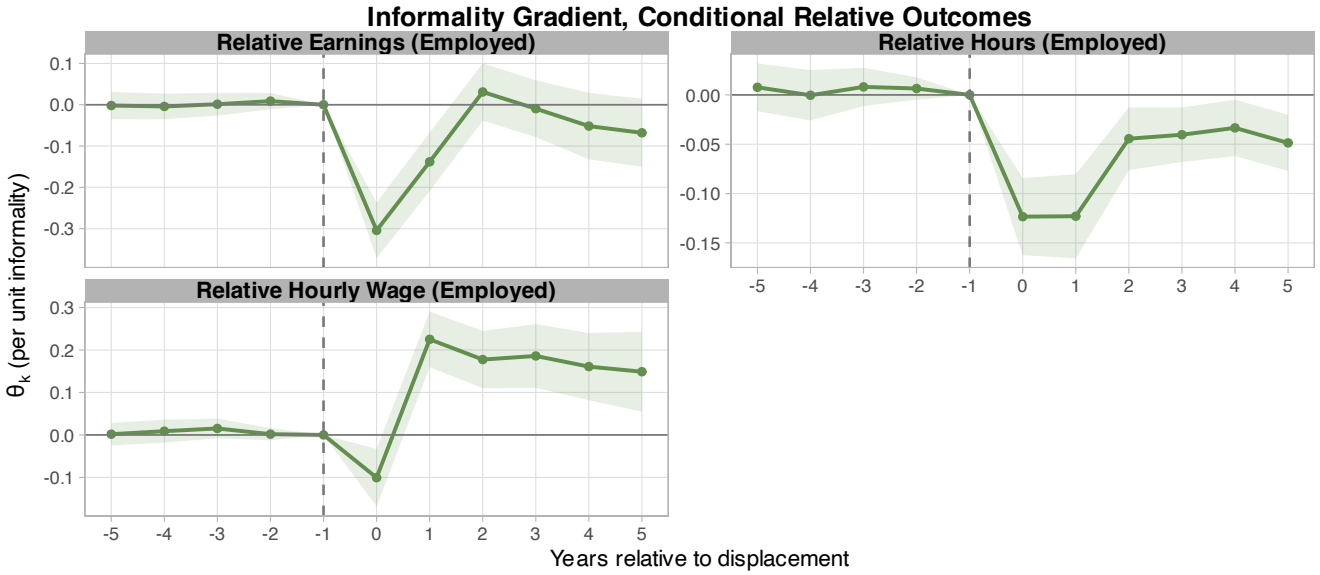


Figure 6: Regional informality gradients conditional on formal employment

Notes: Earnings and hours outcomes require positive formal earnings in the year. The hourly-wage ratio additionally requires valid positive hours and benchmarks. This restriction permits partial-year formal employment. Estimation and confidence intervals are as in Figure 5.

as complementary evidence rather than treated as interchangeable estimates.

Conditioning on positive formal earnings changes the persistence of the earnings gradient (Figure 6). It is negative and statistically significant in the displacement year and year one, at −0.304 and −0.138. These coefficients imply additional losses of 3.0 and 1.4 percentage points of the benchmark for a 10 percentage point increase in informality. From year two onward, the conditional earnings gradients are not statistically distinguishable from zero. This is evidence that the initial additional loss extends to workers with some formal work during the year; it does not establish that later differences disappear. Conditional hours retain a negative, significant gradient through year five, when $\theta_5 = -0.049$.

The wage results again differ from those for total formal earnings and hours. The log hourly-wage gradient is positive from year one onward, at 0.258 in year one and 0.222 in year five (Figure 7). A 10 percentage point increase in informality is associated with a wage effect that is approximately 0.026 log points more positive in year one and 0.022 log points more positive in year five. Log earnings also have a positive gradient at these horizons, while the log-hours gradient is not statistically different from zero. These results coexist with the negative gradients in relative earnings and hours: the log and benchmark-relative outcomes place different emphasis on the distribution, and the latter include years with zero formal earnings. Separate outcome-specific weights also mean that the second-step

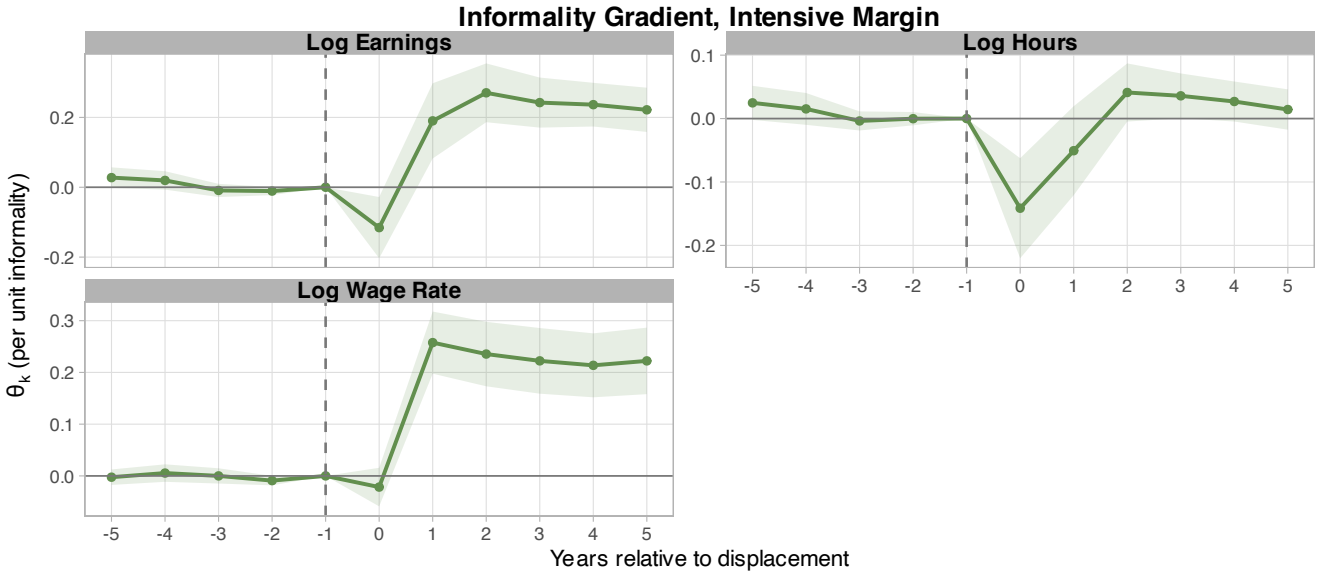


Figure 7: Regional informality gradients in log outcomes

Notes: First-step log outcomes exclude zero or invalid observations. Each second-step regression uses the precision weights for its own outcome and horizon. Estimation and confidence intervals otherwise follow Figure 5.

coefficients do not form an exact additive decomposition of earnings into wages and hours.

3.4 Non-return and subsequent displacement

The event studies describe workers who return within five years. To assess the selection underlying these estimates, I also examine non-return in the broader cohort of eligible displaced workers. Figure 8 shows a positive relationship between regional informality and the share never observed in a later year of formal employment. Across the 527 regions with at least 100 eligible displaced workers, the Spearman rank correlation is 0.63. This measure follows workers through the end of the available records; it is distinct from non-return within a common five-year window. Absence from RAIS identifies an absence from observed formal employment, not its destination.

Evidence on recurrent job loss and models of job security motivate examining whether formal reemployment is followed by further displacement (Stevens, 1997; Jarosch, 2023). Subsequent displacement is common among returners in this sample. In the baseline event study, the estimated difference in annual displacement incidence rises from 7.3 percentage points in year one to 14.1 percentage points in year four. The quartile comparison shows higher subsequent-displacement effects in low-informality regions: in year five, the coefficients are 13.8 percentage points in the lowest quartile and 10.1 in the highest. The continuous second-step gradient is negative as well. This contrasts with the larger formal-employment losses in high-informality regions in the quartile comparison. The

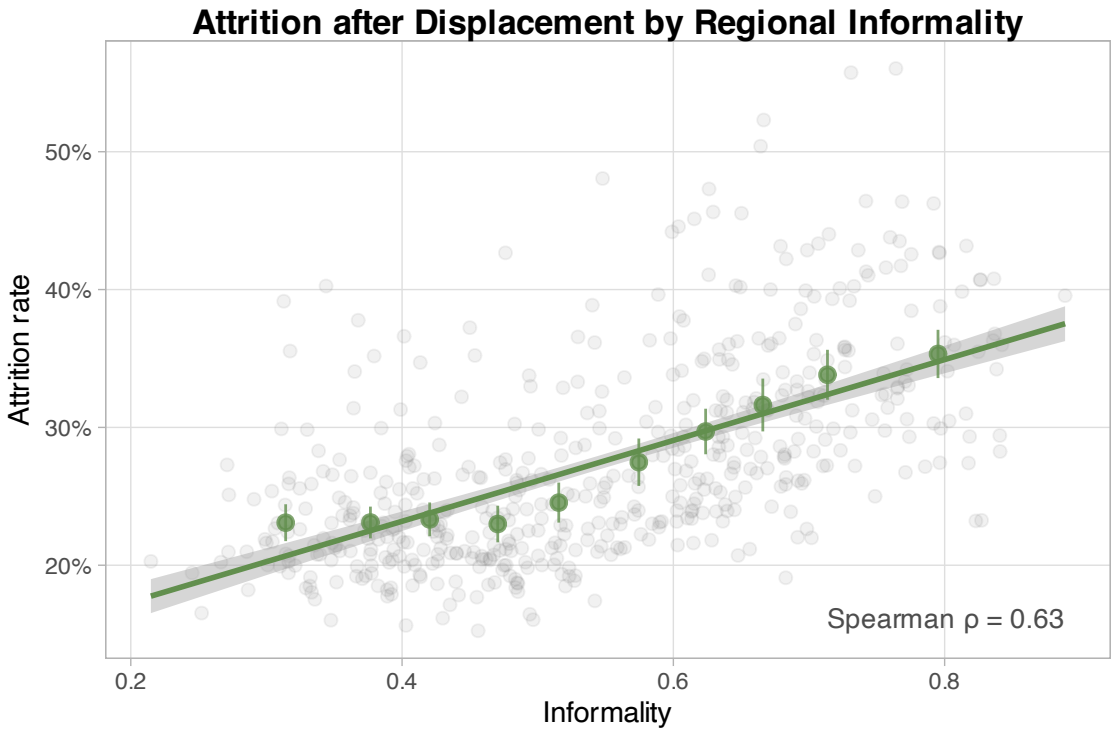


Figure 8: Regional informality and non-return to observed formal employment

Notes: The figure relates regional informality to the share of eligible displaced workers with no formal-employment record in a year after their first qualifying displacement, through the end of observation. It uses the broader displaced cohort, before the five-year-returner restriction, and retains regions with at least 100 eligible displaced workers. Follow-up length can differ across displacement cohorts.

second-step formal-employment gradient, however, is not statistically distinguishable from zero at the 5% level in year one or year five.

Figure 9 shows the employment and subsequent displacement paths together. Greater formal-employment exposure can create more opportunities for another displacement. A higher annual displacement incidence therefore need not imply a greater separation risk conditional on holding a formal job. Moreover, never-displaced workers have zero displacement by construction. This outcome is informative about recurrent displacement among treated workers, but does not by itself identify the effect of the first displacement on a subsequent separation hazard.

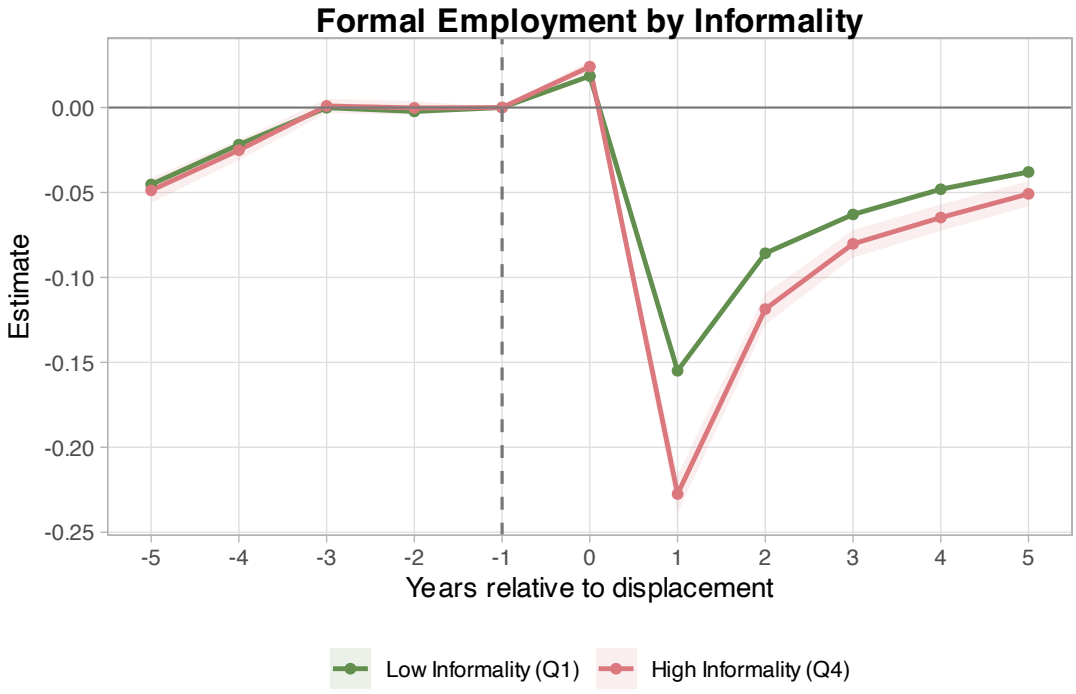
3.5 Interpretation

The results point to substantial and persistent disruption of formal-sector work after displacement. Higher regional informality is associated with larger initial losses and, in the continuous regional analysis, persistent additional losses in earnings and hours relative to pre-displacement benchmarks. Yet workers observed in formal employment in higher-informality regions experience smaller hourly-wage penalties. Formal earnings, contracted hours, and hourly pay consequently provide different views of post-displacement adjustment.

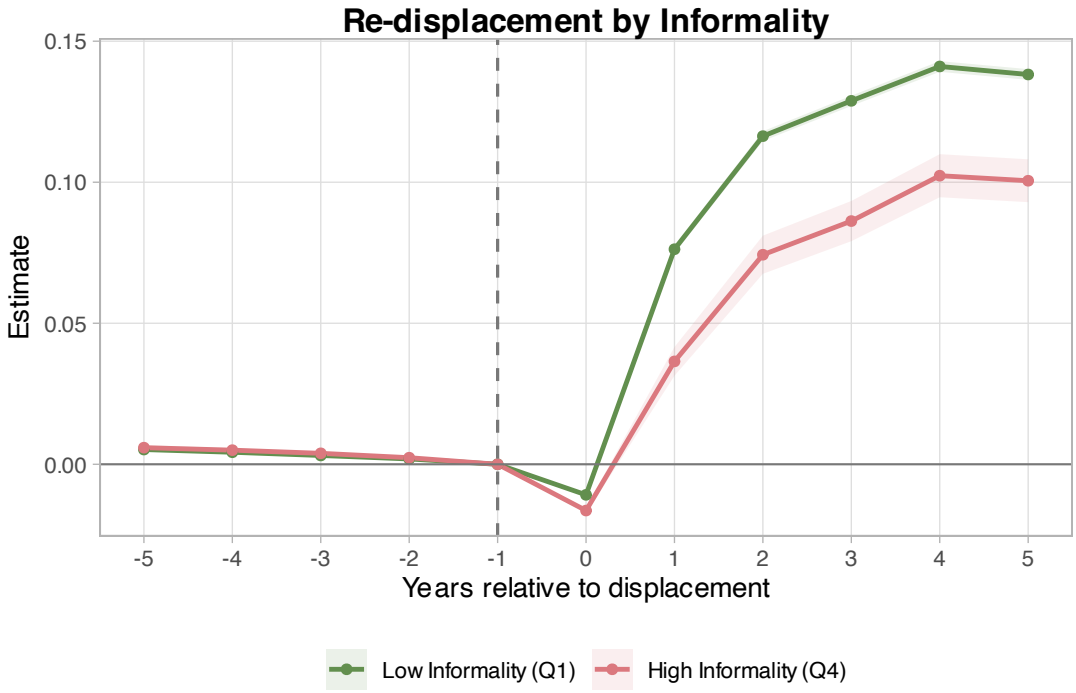
One possible explanation for the wage pattern is that opportunities for informal work improve the alternatives available to a displaced worker, allowing greater selectivity over the wage offered upon formal re-employment. The same setting could support longer spells outside formal work or fewer formal hours. The observed combination is consistent with this possibility, but it does not identify that mechanism. In particular, the data do not show whether workers outside RAIS are earning income informally, so larger formal earnings losses cannot be equated with larger losses in total labour income.

Selection offers another explanation for the wage pattern. In regions where fewer displaced workers return, the returners may have different formal-sector prospects from those who remain outside. Conditioning again on positive earnings in a particular year further changes the group whose wages are observed. The direction of the resulting selection is not determined by the present estimates. Accordingly, the positive wage gradient should not be read as evidence that informal employment protects every displaced worker's wage.

Finally, regional informality is associated with other features of local labour markets, including worker education and market size. The current regional specification does not distinguish infor-



(a) Formal employment



(b) Subsequent displacement

Figure 9: Formal employment and subsequent displacement by regional informality

Notes: Groups and confidence intervals are as in Figure 4. Formal employment requires positive earnings at some point in the year. Subsequent displacement measures annual incidence after the initial event, rather than cumulative incidence or a hazard conditional on formal employment. Never-displaced comparison workers have zero displacement by construction.

mality from those possible explanations for differences in displacement losses. Together with the pre-period differences, these considerations limit a causal interpretation of the regional gradients. The evidence establishes differences in observed formal-sector adjustment; explaining their sources requires additional comparisons of regional characteristics, employment exposure, and selection into formal re-employment.

4 Model

I use a two-sector extension of the sequential search framework of McCall (1970) to examine how informal job opportunities affect workers' acceptance of formal jobs after displacement. The model distinguishes the wage at which a worker accepts formal employment from the time taken to return to that sector. It provides a qualitative framework for interpreting these margins; the comparative statics hold wage offer distributions and other labor-market conditions fixed.

Job ladders and employment stability are central to quantitative search explanations of displacement losses (Krolikowski, 2017; Jung and Kuhn, 2019; Jarosch, 2023). Meghir et al. (2015) model formal and informal employment jointly in equilibrium. The exercise here isolates how access to informal offers changes formal-job acceptance, taking firms' decisions and the opportunities available in each sector as given.

4.1 Environment

Time is continuous. Workers are risk neutral, discount future income at rate $r > 0$, and may be unemployed (U), formally employed (F), or informally employed (I). Search is undirected, and the environment is stationary and partial equilibrium: workers take offer distributions, arrival rates, and separation rates as given.

An unemployed worker receives flow income b. Formal and informal job offers arrive according to independent Poisson processes with rates λ_F and λ_I. An offer in sector $s \in \{F, I\}$ pays a constant wage drawn independently from the stationary distribution G_s, with support Ω_s. Rejected offers cannot be recalled. For the derivations, I assume nonnegative wages with finite first moments and distributions that are continuously differentiable near the reservation wage.

An accepted job ends at the exogenous, wage-independent rate $\delta_s > 0$, returning the worker to unemployment. Workers do not search while employed; movement between sectors therefore

requires an intervening unemployment spell.

Let U denote the value of unemployment and $V_s(w)$ the value of employment in sector s at wage w. These values satisfy

$$rV_s(w) = w + \delta_s\big[U - V_s(w)\big], \qquad s \in \{F, I\}, \tag{4}$$

$$rU = b + \sum_{s\in\{F,I\}} \lambda_s \int_{\Omega_s} \max\{V_s(w) - U, 0\}\, dG_s(w). \tag{5}$$

Employment yields the wage until separation, while unemployment yields b and the option to accept sufficiently valuable offers from either sector.

4.2 Informal opportunities and job acceptance

An offer is accepted when $V_s(w) \geq U$. At the reservation wage, the worker is indifferent between employment and continued search. Equation (4) then implies a common cutoff $w^* = rU$ in both sectors, even when their separation rates differ. I therefore omit the sector subscript on the reservation wage. Substituting the job values into equation (5) gives

$$w^* = b + \sum_{s\in\{F,I\}} \frac{\lambda_s}{r + \delta_s} \int_{w\geq w^*} (w - w^*)\, dG_s(w). \tag{6}$$

The cutoff equals current unemployment income plus the option value of future job offers. Holding other primitives fixed, a higher λ_I weakly raises U and w^*, strictly so when informal offers above the cutoff occur with positive probability. Better access to informal opportunities therefore makes workers more selective about formal jobs as well as informal jobs.

Offer arrivals translate into accepted-job transition rates

$$h_F = \lambda_F\big[1 - G_F(w^*)\big], \qquad h_I = \lambda_I\big[1 - G_I(w^*)\big]. \tag{7}$$

With λ_F and G_F fixed, the higher cutoff weakly lowers the rate at which unemployed workers enter formal employment and weakly raises the mean accepted formal wage, $E[w_F \mid w_F \geq w^*]$. This wage change reflects greater selectivity over the same offer distribution. The effect on h_I is ambiguous: informal offers arrive more frequently, but a smaller fraction is accepted. Appendix A derives these results.

To isolate this mechanism, I consider variation in λ_I alone. A higher regional informality share need not correspond to a change in this single parameter: formal opportunities, separation rates, and offer distributions can also differ across regions. In particular, a simultaneous reduction in λ_F would lower the reservation wage and could offset the effect of higher λ_I.

4.3 Time to formal return

The expected duration of an unemployment spell is $1/(h_F + h_I)$, since either type of accepted job ends unemployment. Returning to formal employment can take longer: a worker may pass through several informal jobs and unemployment spells before accepting a formal offer. Let D_U be the expected time to first formal employment starting from unemployment. For $h_F > 0$ and $\delta_I > 0$,

$$D_U = \frac{1 + h_I/\delta_I}{h_F} = \frac{1}{h_F} + \frac{h_I}{h_F}\frac{1}{\delta_I}. \tag{8}$$

The first term is expected total time unemployed before formal return; the second is expected time in informal employment. Informal spells interrupt formal search under the assumption of no job offers while employed.

At a fixed cutoff, an increase in λ_I raises D_U whenever informal offers have a positive probability of acceptance: it increases entry into informal work and the time spent there before formal return. The induced increase in the cutoff has competing effects. Rejecting more formal offers delays return, while rejecting more informal offers avoids interruptions to formal search. Consequently, the overall effect on D_U cannot be signed without further restrictions. Appendix A gives the derivative and a sufficient condition for a positive effect, including the illustrative case of identical formal and informal offer distributions. That restriction is not imposed in the baseline model.

4.4 Connection to the empirical results

The model highlights a distinction between formal-job acceptance and wages among workers who return. More frequent informal opportunities can raise the pay of accepted formal jobs while reducing formal entry from unemployment. This provides a mechanism to consider alongside the smaller conditional wage penalties and weaker formal-employment outcomes observed in higher-informality regions. It does not establish that formal return must take longer, or that formal earnings losses must be larger: time in formal work and pay while employed can move in opposite directions,

and their combined effect is not determined here.

The wage comparison also requires care. The model predicts a higher level of accepted wages when λ_I increases; the event studies estimate wage changes relative to a comparison trajectory. For a given wage on the lost job, higher reemployment wages would imply a smaller wage loss, but the model does not establish that pre-displacement wages are comparable across regions. In a stationary version of this environment with wage-independent separations, old and newly accepted formal wages follow the same truncated offer distribution, so the baseline does not by itself generate average wage scarring.

Finally, D_U is an unrestricted expectation, whereas the event-study sample conditions on formal return within five years. RAIS also does not distinguish informal employment from unemployment or nonparticipation during periods outside formal work. The model therefore supplies a qualitative account of job acceptance, rather than a quantitative mapping from regional informality to the estimated displacement effects or to losses in total labor income.

5 Conclusion

This paper examines how recovery after job displacement varies with regional informality in Brazil. Among displaced workers who return to formal employment within five years, formal earnings and hours remain below the comparison trajectory well after displacement. Higher-informality regions exhibit larger initial earnings and employment losses, alongside smaller hourly-wage penalties among workers with observed positive formal earnings and hours. The persistence of regional differences depends on the outcome and specification. These patterns underscore the value of examining formal employment, hours, and hourly pay separately when describing post-displacement adjustment.

The search model provides one interpretation of the opportunities facing displaced workers. More frequent informal offers can raise the reservation wage and the mean accepted formal wage, holding other labor-market conditions fixed. The effect on time to formal return remains ambiguous because accepting an informal job interrupts formal search, while greater selectivity reduces acceptance of both types of offers. This mechanism offers a way to organize the empirical patterns, but selection into reemployment and differences in formal job opportunities also remain possible explanations for the regional wage comparisons.

The empirical findings are preliminary. Differences in worker and sector composition, including the prevalence of public administration among controls, and differences in pre-displacement trends warrant further work on the comparison group. The five-year-return restriction also selects the workers whose recovery is measured. Moreover, RAIS does not reveal workers' activities or earnings outside formal employment. Complementary evidence on transitions among formal work, informal work, unemployment, and nonparticipation would help distinguish the mechanisms behind the observed patterns and extend the analysis beyond formal-sector outcomes.

A further direction is to examine how policies aimed at reducing informality affect recovery after displacement and overall welfare. Policies that expand formal opportunities and policies that restrict informal employment can alter different margins of workers' options and firms' decisions. Evaluating these changes would require evidence on earnings outside formal employment and a framework that accounts for responses in labor demand, wages, and sector choice. Assessing welfare would also require accounting for workers' ability to insure against earnings risk (Rogerson and Schindler, 2002). A quantitative equilibrium model of post-displacement adjustment could examine how the effects on formal recovery, total income, and welfare vary across workers and policy instruments. The present analysis provides evidence on formal-sector adjustment that could inform such an exercise, while leaving the welfare effects of these policies open.

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A Theoretical Appendix

A.1 Reservation wages and comparative statics

Solving equation (4) for the value of a job gives

$$V_s(w) = \frac{w + \delta_s U}{r + \delta_s}, \qquad V_s(w) - U = \frac{w - rU}{r + \delta_s}. \tag{A.1}$$

The sign of the surplus is therefore the same in both sectors at any given wage. The common acceptance cutoff is $w^* = rU$. Substitution into the unemployment Bellman equation yields equation (6).

Define the expected excess wage and the acceptance probability at cutoff c as

$$A_s(c) = \int_{w \geq c} (w - c)\, dG_s(w), \qquad S_s(c) = 1 - G_s(c). \tag{A.2}$$

Finite first moments ensure that $A_s(c)$ is finite. Where G_s is continuous, Leibniz's rule gives $A_s'(c) = -S_s(c)$: the boundary term is zero because an offer at the cutoff has zero surplus. The cutoff solves

$$H(c) = c - b - \sum_{s \in \{F,I\}} \frac{\lambda_s}{r + \delta_s} A_s(c) = 0. \tag{A.3}$$

Suppressing the other arguments of H, its relevant partial derivatives are

$$H_c = 1 + \sum_{s \in \{F,I\}} \frac{\lambda_s S_s(c)}{r + \delta_s} > 0, \tag{A.4}$$

$$H_{\lambda_s} = -\frac{A_s(c)}{r + \delta_s} \leq 0, \qquad H_b = -1. \tag{A.5}$$

The reservation-wage equation thus has at most one solution. At that solution, the implicit function theorem implies

$$\frac{\partial w^*}{\partial \lambda_s} = \frac{A_s(w^*)}{(r + \delta_s) H_c(w^*)} \geq 0, \qquad \frac{\partial w^*}{\partial b} = \frac{1}{H_c(w^*)} > 0. \tag{A.6}$$

The arrival-rate effect is strictly positive when offers strictly above the cutoff have positive probability. These are partial-equilibrium comparisons holding the remaining primitives fixed.

A.2 Accepted wages and transition rates

Let $\mu_F(c) = E[w_F \mid w_F \geq c]$, for $S_F(c) > 0$. Under the smoothness conditions in the main text,

$$\mu_F'(c) = \frac{g_F(c)}{S_F(c)}\big[\mu_F(c) - c\big] \geq 0. \tag{A.7}$$

Consequently, the increase in w^* induced by higher λ_I weakly raises the mean accepted formal wage. With $g_s = G_s'(w^*)$ and $S_s = S_s(w^*)$,

$$\frac{dh_F}{d\lambda_I} = -\lambda_F g_F \frac{\partial w^*}{\partial \lambda_I} \leq 0, \tag{A.8}$$

$$\frac{dh_I}{d\lambda_I} = S_I - \lambda_I g_I \frac{\partial w^*}{\partial \lambda_I}. \tag{A.9}$$

The informal entry rate combines a positive direct effect with a negative acceptance effect. The corresponding response of the total unemployment exit rate is $S_I - (\lambda_F g_F + \lambda_I g_I)\partial w^*/\partial \lambda_I$; its sign, and therefore the sign of the response of $1/(h_F + h_I)$, is not determined by the increase in the cutoff alone.

A.3 Expected time to formal employment

Let D_U and D_I be the expected times to first formal employment starting from unemployment and informal employment, respectively. Assume $h_F > 0$ and $\delta_I > 0$. The next accepted job arrives after an expected $1/(h_F + h_I)$ units of time. It is informal with probability $h_I/(h_F + h_I)$; a formal arrival ends the waiting time. An informal spell lasts an expected $1/\delta_I$, after which the worker faces the same unemployment problem. Thus

$$D_U = \frac{1}{h_F + h_I} + \frac{h_I}{h_F + h_I} D_I, \tag{A.10}$$

$$D_I = \frac{1}{\delta_I} + D_U. \tag{A.11}$$

Solving the two equations yields

$$D_U = \frac{1 + h_I/\delta_I}{h_F}, \qquad D_I = \frac{1}{\delta_I} + \frac{1 + h_I/\delta_I}{h_F}. \tag{A.12}$$

The expected number of informal spells before formal entry is h_I/h_F. Each lasts $1/\delta_I$ on average, giving the decomposition in equation (8).

A.4 The effect of informal offer arrivals on formal return

Write expected duration as a function of arrival rates and the cutoff:

$$\Phi(\lambda_I, \lambda_F, c) = \frac{1 + \lambda_I[1 - G_I(c)]/\delta_I}{\lambda_F[1 - G_F(c)]}. \tag{A.13}$$

Holding λ_F, the offer distributions, and the remaining primitives fixed, the total derivative at $c = w^*$ is

$$\frac{dD_U}{d\lambda_I} = \Phi_{\lambda_I} + \Phi_c \frac{\partial w^*}{\partial \lambda_I}. \tag{A.14}$$

For $S_I, S_F > 0$, the direct effect is

$$\Phi_{\lambda_I} = \frac{S_I}{\delta_I \lambda_F S_F} > 0. \tag{A.15}$$

The cutoff effect follows from the quotient rule:

$$\begin{aligned} \Phi_c &= -\frac{\lambda_I g_I}{\delta_I \lambda_F S_F} + \frac{(1 + \lambda_I S_I/\delta_I) g_F}{\lambda_F S_F^2} \\ &= \Phi \left[\frac{g_F}{S_F} - \frac{h_I}{\delta_I + h_I} \frac{g_I}{S_I} \right]. \end{aligned} \tag{A.16}$$

It is positive precisely when

$$\frac{g_F(w^*)}{1 - G_F(w^*)} > \frac{h_I}{\delta_I + h_I} \frac{g_I(w^*)}{1 - G_I(w^*)}. \tag{A.17}$$

The first term measures the reduction in formal acceptance as the cutoff rises; the second captures the avoidance of informal spells. A nonnegative Φ_c is sufficient, but not necessary, for a positive total duration effect, because $\Phi_{\lambda_I} > 0$. A negative cutoff effect can be outweighed by the direct effect.

As an illustrative sufficient case, suppose $G_F = G_I = G$. Then

$$\Phi_c = \Phi \frac{\delta_I}{\delta_I + h_I} \frac{g(w^*)}{1 - G(w^*)} \geq 0, \tag{A.18}$$

and $dD_U/d\lambda_I > 0$. This example demonstrates a condition under which formal return takes longer; it is not an assumption about the wage distributions in the empirical application.